Egypt Residential Real Estate Power BI

Project Planning & Management

Team: Saleh, Maged, Abd Elkhalek, Sama Raed, Sama Abdelmeged, Mostafa

1. Project Proposal

Overview. Build an interactive Power BI report on Egyptian residential real estate transactions, using a star schema: 1 fact table (about 25,000 transactions) and 8 dimensions (Date, Property, PropertyType, Location, Developer, Agent, Customer, Channel).

Objectives

- Give stakeholders a clear view of sales, pricing, cancellations, and agent/channel performance.
- Deliver a validated, well-modeled dataset with documented metric definitions.
- Publish a polished, navigable report to the agreed workspace.

Scope

- *In scope:* data cleaning, star-schema modeling, DAX measures, report pages, interactivity, validation, publishing, documentation.
- *Out of scope:* live data feeds, predictive models, external data sources.

2. Project Plan

Timeline (Gantt, 6 weeks - adjust to the actual deadline)

Phase	Owner	W1	W2	W3	W4	W5	W6
Requirements & kickoff	Sama Raed						
Data preparation (Power Query)	Saleh						
Data modeling	Sama Raed						
DAX measures & metrics	Maged						
Report design & KPI cards	Abd Elkhalek						
Interactivity & navigation	Mostafa						
Quality review	Maged						
Validation & publishing	Sama Abdelmeged						
Documentation & handover	Sama Abdelmeged						
Final presentation	Abd Elkhalek						

Milestones

- M1 (end W1): requirements approved, data loaded.
- M2 (end W2): clean data ready.
- M3 (end W3): relationships validated.
- M4 (end W4): core measures complete.
- M5 (end W5): full report draft with interactivity.
- M6 (end W6): validated, published, presented.

Deliverables: requirements document, cleaned dataset, data model, DAX measure library, Power BI report (.pbix), validation checklist, metric definitions, handover pack, final presentation.

Resources: Power BI Desktop and Service, Power Query, DAX, the 10 source CSV files, shared workspace/drive, weekly team meeting.

3. Task Assignment & Roles

Member	Technical responsibility	Non-technical role	Key outputs
Saleh	Power Query & data preparation: import, clean, transform; check data types and missing values	Project Coordinator: track tasks, deadlines, progress	Clean tables, data-quality log, task tracker
Maged	DAX & metrics: measures and calculated columns for key metrics and trends	Quality Reviewer: check clarity, consistency, ease of understanding	Measure library, review feedback list
Abd Elkhalek	Report design: pages, visuals, KPI cards	Team Leader & Presenter: coordinate team, present findings	Report pages, final presentation
Sama Raed	Data modeling: create and validate fact-dimension relationships	Requirements Lead: gather and organize stakeholder needs	Requirements doc, star-schema model
Sama Abdelmeged	Report validation & publishing: verify measures/visuals, publish to workspace	Documentation Lead: project notes, metric definitions, handover	Validation checklist, published report, handover pack
Mostafa	Interactivity & navigation: slicers, filters, drill-through, navigation	Meeting Facilitator: organize discussions, record action items	Interactive report, meeting minutes

Handoff flow: Sama Raed (requirements) -> Saleh (data) -> Sama Raed (model) -> Maged (DAX) -> Abd Elkhalek and Mostafa (design and interactivity) -> Maged (review) -> Sama Abdelmeged (validate and publish) -> Abd Elkhalek (present).

4. Risk Assessment & Mitigation Plan

#	Risk	Likelihood	Impact	Mitigation	Owner
1	Data quality issues (nulls, wrong types, duplicates)	Medium	High	Profile data early; keep a data-quality log; use the validation report	Saleh
2	Wrong or broken relationships (duplicated keys, ambiguous filters)	Medium	High	Validate cardinality and filter direction; test with sample totals	Sama Raed
3	Incorrect DAX (e.g., mixing cancelled and successful deals)	Medium	High	Document each definition; cross-check vs source totals; peer review	Maged
4	Slow report performance with 25K rows and many visuals	Low	Medium	Limit visuals per page; prefer measures over calculated columns; remove unused columns	Abd Elkhalek
5	Cluttered or inconsistent design	Medium	Medium	Agree on a theme and layout template; quality review before validation	Maged / Abd Elkhalek
6	Broken slicers, drill-through, or navigation	Medium	Medium	Test every interaction on each page; keep a test checklist	Mostafa
7	Requirements change or are unclear	Medium	Medium	Sign off requirements in W1; log changes with approval	Sama Raed
8	Missed deadlines or unclear task ownership	Medium	High	Weekly tracker review; early escalation of blockers	Saleh
9	Publishing or workspace access problems	Low	High	Confirm workspace access in W2; trial publish in W4	Sama Abdelmeged
10	Missing or outdated documentation	Medium	Medium	Update docs weekly; finalize metric definitions at M4	Sama Abdelmeged
11	Poor communication and unrecorded decisions	Low	Medium	Weekly meeting with minutes and action items	Mostafa

5. KPIs

A. Project success KPIs

KPI	Target	Owner
Milestones delivered on time	>= 90%	Saleh
Tasks completed vs planned (weekly)	>= 90%	Saleh
Data quality: null/invalid values after cleaning	0 in key columns	Saleh
Relationship validation: tables correctly related	100%	Sama Raed
Measure accuracy vs source totals	100% match	Sama Abdelmeged
Report page load / visual response time	< 5 seconds	Abd Elkhalek
Interaction test pass rate (slicers, drill-through, navigation)	100%	Mostafa
Quality review issues closed before publishing	100%	Maged
Meeting action items closed on time	>= 85%	Mostafa
Report availability in workspace (uptime)	>= 99%	Sama Abdelmeged
Stakeholder user adoption (viewers within first month)	>= 80% of target users	Abd Elkhalek
Stakeholder satisfaction (clarity and usefulness)	>= 4 / 5	Sama Raed

B. Report business KPIs (built by Maged, displayed by Abd Elkhalek)

- Total transactions and total net sales (EGP)
- Average price per sqm (EGP)
- Successful sale rate and cancellation rate
- Average days to close
- Average discount rate
- Total commission (EGP)
- Sales by location, property type, developer, agent, and channel
- Monthly and yearly sales trend

6. Summary

Six members deliver a Power BI real estate report in about 6 weeks. Saleh prepares data and coordinates, Sama Raed defines requirements and the model, Maged builds the DAX and reviews quality, Abd Elkhalek designs and presents, Mostafa adds interactivity and runs meetings, and Sama Abdelmeged validates, publishes, and documents. Top risks are data quality, model and DAX errors, and deadlines, each with a named owner. Success is tracked through on-time delivery, 100% measure accuracy, fast response time, high uptime, and strong user adoption.

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Literature Review: Feedback, Improvements & Grading

Team: Saleh, Maged, Abd Elkhalek, Sama Raed, Sama Abdelmeged, Mostafa

1. Feedback & Evaluation

Purpose. Record the lecturer's assessment of the project and map each comment to the team member responsible. Fields marked *Lecturer* are left open until feedback is received. The *Team self-check* column is the team's own pre-review assessment.

Evidence from the project (validation report): all 11 data checks passed, including 25,000 unique transactions, full key coverage for property, location, and date, correct amount and balance logic, zero net amount for cancelled deals, and valid Egyptian geography.

Area	Team self-check	Lecturer's assessment	Responsible
Data preparation	Strong: all validation checks passed	*Lecturer*	Saleh
Data model	Star schema with 1 fact and 8 dimensions; relationships validated	*Lecturer*	Sama Raed
DAX & metrics	Core KPIs defined; definitions documented	*Lecturer*	Maged
Report design	Clear KPI cards and consistent theme	*Lecturer*	Abd Elkhalek
Interactivity	Slicers, drill-through, and navigation tested	*Lecturer*	Mostafa
Validation & publishing	Totals checked against source; report published	*Lecturer*	Sama Abdelmeged
Requirements & scope	Stakeholder needs documented	*Lecturer*	Sama Raed
Project management	Milestones and tasks tracked	*Lecturer*	Saleh
Documentation	Metric definitions and handover pack	*Lecturer*	Sama Abdelmeged
Presentation	Findings presented clearly	*Lecturer*	Abd Elkhalek

Overall comment (Lecturer): _____

2. Suggested Improvements

#	Improvement	Benefit	Owner
1	Add automated data-quality checks in Power Query (null, duplicate, and range rules that run on every refresh)	Problems are caught at refresh, not by manual review	Saleh
2	Add a formal data dictionary covering every table and column	Easier onboarding and handover	Sama Abdelmeged
3	Add time intelligence measures (year-over-year, month-over-month, running totals)	Stronger trend analysis	Maged
4	Add a measure-testing table that compares each DAX result with a source total	Proves measure accuracy	Maged
5	Add an executive summary page and a drill-down page for locations and developers	Faster insight for managers	Abd Elkhalek
6	Add bookmarks, tooltips, and a reset-filters button	Better usability	Mostafa
7	Add accessibility checks (contrast, alt text, tab order) and mobile layout	Wider audience	Abd Elkhalek
8	Add row-level security by region or agent office if required	Controlled access to data	Sama Abdelmeged
9	Set up scheduled refresh and report usage monitoring	Supports uptime and adoption KPIs	Sama Abdelmeged
10	Run a short user feedback session and log changes	Evidence-based improvement	Sama Raed
11	Keep meeting minutes and a change log in one shared place	Traceable decisions	Mostafa
12	Run a peer review checklist before each release	Consistent quality	Maged

Priority: high = 3, 4, 9; medium = 1, 2, 5, 8; low = 6, 7, 10, 11, 12.

3. Final Grading Criteria

Proposed breakdown. Replace with the official course rubric if it differs.

Category	Marks	What is assessed	Main contributors
Documentation	25	Proposal, plan, roles, risks, KPIs, metric definitions, handover	Sama Abdelmeged, Sama Raed, Saleh
Implementation	40	Data preparation, star-schema model, DAX measures, report design, interactivity	Saleh, Sama Raed, Maged, Abd Elkhalek, Mostafa
Testing	15	Data validation, measure accuracy, interaction tests, review and publishing checks	Sama Abdelmeged, Maged, Mostafa
Presentation	20	Clarity, structure, demo of the report, answers to questions	Abd Elkhalek (lead), whole team
Total	**100**		

Detailed marking guide

Category	Criterion	Marks
Documentation	Completeness and structure	10
Documentation	Metric definitions and data dictionary	8
Documentation	Clarity of roles, risks, and KPIs	7
Implementation	Data cleaning and preparation	8
Implementation	Data model quality	8
Implementation	DAX measures and accuracy	10
Implementation	Report design and KPI cards	8
Implementation	Interactivity and navigation	6
Testing	Data and relationship validation	5
Testing	Measure and visual verification	5
Testing	Interaction tests and publishing check	5
Presentation	Content and storytelling	8
Presentation	Demo and delivery	7
Presentation	Q&A and teamwork	5

Mark record

Category	Marks available	Marks awarded	Lecturer comment
Documentation	25		
Implementation	40		
Testing	15		
Presentation	20		
Total	**100**		

4. Summary

The project's data checks all pass and each member owns a clear area. The lecturer's assessment is left open for entry. The 12 suggested improvements focus on measure testing, time intelligence, scheduled refresh, and documentation, and each has an owner. Marks are proposed as Documentation 25, Implementation 40, Testing 15, and Presentation 20, and each category is linked to the members who deliver it.

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Requirements Gathering

Team: Saleh, Maged, Abd Elkhalek, Sama Raed, Sama Abdelmeged, Mostafa. Requirements Lead: Sama Raed.

1. Stakeholder Analysis

Stakeholder	Interest	Key needs	Influence	Team contact
Sales / executive managers	Overall sales performance	Total net sales, trends, cancellation rate, top locations and developers	High	Abd Elkhalek
Sales team leaders	Agent and office performance	Sales by agent, agent level, office region, and channel; commission totals	High	Maged
Agents	Own results and targets	Own sales, discounts, days to close	Medium	Mostafa
Marketing / channel managers	Channel effectiveness	Sales and commission rate by channel; customer segment and nationality mix	Medium	Sama Raed
Pricing / market analysts	Price levels and demand	Price per sqm by location, market tier, property type, finishing level	Medium	Maged
Developer relations	Developer performance	Sales and cancellations by developer and developer tier	Medium	Sama Raed
Finance team	Revenue and receivables	Gross vs net amount, down payment, balance due, payment method	Medium	Sama Abdelmeged
Lecturer / evaluators	Project quality	Complete documentation, correct metrics, clear presentation	High	Abd Elkhalek
Project team	Delivery	Clear scope, clean data, defined metrics	High	Saleh

Gathering methods: stakeholder interviews and short surveys (Sama Raed), review of the 10 source files and validation report (Saleh), requirements workshop with sign-off in week 1 (Sama Raed, notes by Mostafa), and a feedback session before publishing.

2. User Stories & Use Cases

User stories

ID	As a...	I want...	So that...	Priority
US-01	Sales manager	a summary page with total net sales, transactions, and cancellation rate	I see overall performance at once	High
US-02	Sales manager	to compare sales by governorate, city, and district	I find the strongest and weakest markets	High
US-03	Team leader	to rank agents by sales and commission	I recognize top performers	High
US-04	Agent	to see my own sales, discounts, and days to close	I track my targets	Medium
US-05	Marketing manager	to compare channels by sales and commission rate	I invest in the best channels	Medium
US-06	Pricing analyst	to view average price per sqm by location, property type, and finishing level	I judge price levels	High
US-07	Developer relations	to see sales and cancellations by developer	I assess partners	Medium
US-08	Finance officer	to see gross amount, discount, net amount, down payment, and balance due	I follow revenue and receivables	Medium
US-09	Any user	to filter by year, month, location, property type, and status	I focus on what matters to me	High
US-10	Any user	to drill from a summary to transaction details	I understand the reason behind a number	Medium
US-11	Any user	to navigate pages with clear buttons and a visible reset-filters option	I use the report without training	Medium
US-12	Lecturer	to read clear metric definitions and the project documentation	I assess the work fairly	High

Use cases

ID	Use case	Actor	Main flow	Result
UC-01	View sales overview	Sales manager	Open report; summary page loads; select a year with the slicer; KPI cards update	Manager sees filtered KPIs
UC-02	Analyze a market	Sales manager	Choose a governorate; map and charts update; drill through to district and property type	Manager identifies top districts
UC-03	Review agent performance	Team leader	Open agent page; filter by office region and agent level; sort by sales	Ranked agent list with commission
UC-04	Compare channels	Marketing manager	Open channel page; compare sales and commission rate; filter by customer segment	Best channel identified
UC-05	Check finance position	Finance officer	Open finance page; filter by payment method; view net amount and balance due	Receivables visible
UC-06	Investigate cancellations	Any user	Select Cancelled status; drill through to transactions list	Cancellation causes explored
UC-07	Refresh and publish	Sama Abdelmeged	Refresh data; run validation checks; publish to workspace	Updated report available

3. Functional Requirements

ID	Requirement	Linked stories	Owner	Priority
FR-01	Import the fact table and all 8 dimension tables and clean data types and missing values	All	Saleh	High
FR-02	Model a star schema with one-to-many relationships from each dimension to the fact table	All	Sama Raed	High
FR-03	Provide measures for total transactions, gross amount, net amount, average price per sqm, average discount, down payment, balance due, and commission	US-01, US-03, US-08	Maged	High
FR-04	Provide successful sale rate, cancellation rate, and average days to close	US-01, US-10	Maged	High
FR-05	Provide time intelligence: year-over-year and month-over-month change, running total	US-01	Maged	Medium
FR-06	Create report pages: overview, locations, property types, developers, agents, channels, finance, details	US-01 to US-08	Abd Elkhalek	High
FR-07	Display KPI cards, trend lines, bar charts, map, and tables with a consistent theme	US-01, US-02	Abd Elkhalek	High
FR-08	Provide slicers for year, month, location, property type, developer, channel, and transaction status	US-09	Mostafa	High
FR-09	Provide drill-through from summary visuals to transaction details	US-10	Mostafa	Medium
FR-10	Provide navigation buttons, bookmarks, and a reset-filters button	US-11	Mostafa	Medium
FR-11	Verify every measure against source totals before release	All	Sama Abdelmeged	High
FR-12	Publish the report to the agreed workspace	UC-07	Sama Abdelmeged	High
FR-13	Provide metric definitions, data dictionary, and handover notes	US-12	Sama Abdelmeged	High
FR-14	Review the report for clarity and consistency before validation	US-11, US-12	Maged	Medium
FR-15	Record requirements, changes, and meeting action items	All	Sama Raed / Mostafa	Medium

4. Non-functional Requirements

ID	Category	Requirement	Measure	Owner
NFR-01	Performance	Report pages load and respond quickly with 25,000 transactions	Under 5 seconds per page and visual update	Abd Elkhalek
NFR-02	Performance	Efficient model: measures over calculated columns, unused columns removed	Model size and refresh time tracked	Saleh / Maged
NFR-03	Security	Only authorized users can open the report	Workspace access limited to approved users	Sama Abdelmeged
NFR-04	Security	Row-level security by region or office when required	Tested with sample roles	Sama Abdelmeged
NFR-05	Security	Customer data is limited to what the analysis needs	Sensitive fields reviewed before publishing	Saleh
NFR-06	Usability	Clear layout, readable fonts, consistent colors, and tooltips	Quality review checklist passed	Maged
NFR-07	Usability	New users can find key figures without training	At least 4 of 5 in a short user test	Sama Raed
NFR-08	Usability	Accessible design: good contrast, alt text, tab order	Accessibility check passed	Abd Elkhalek
NFR-09	Reliability	Report is available in the workspace	Availability of at least 99%	Sama Abdelmeged
NFR-10	Reliability	Data refresh completes without errors	Scheduled refresh succeeds	Saleh
NFR-11	Reliability	Measures return accurate results	100% match with source totals	Maged
NFR-12	Maintainability	Measures and tables are named and documented	Data dictionary complete	Sama Abdelmeged

5. Summary

Nine stakeholder groups need clear sales, price, agent, channel, developer, and finance views from 25,000 transactions. Twelve user stories and seven use cases describe how they work with the report. Fifteen functional requirements cover data preparation, modeling, measures, report pages, interactivity, validation, publishing, and documentation, each with a named owner. Twelve non-functional requirements set targets for performance (under 5 seconds), security, usability, and reliability (99% availability, 100% measure accuracy).

Egypt Residential Real Estate Power BI

System Analysis & Design: Database Design & Data Modeling, UI/UX Design & Prototyping

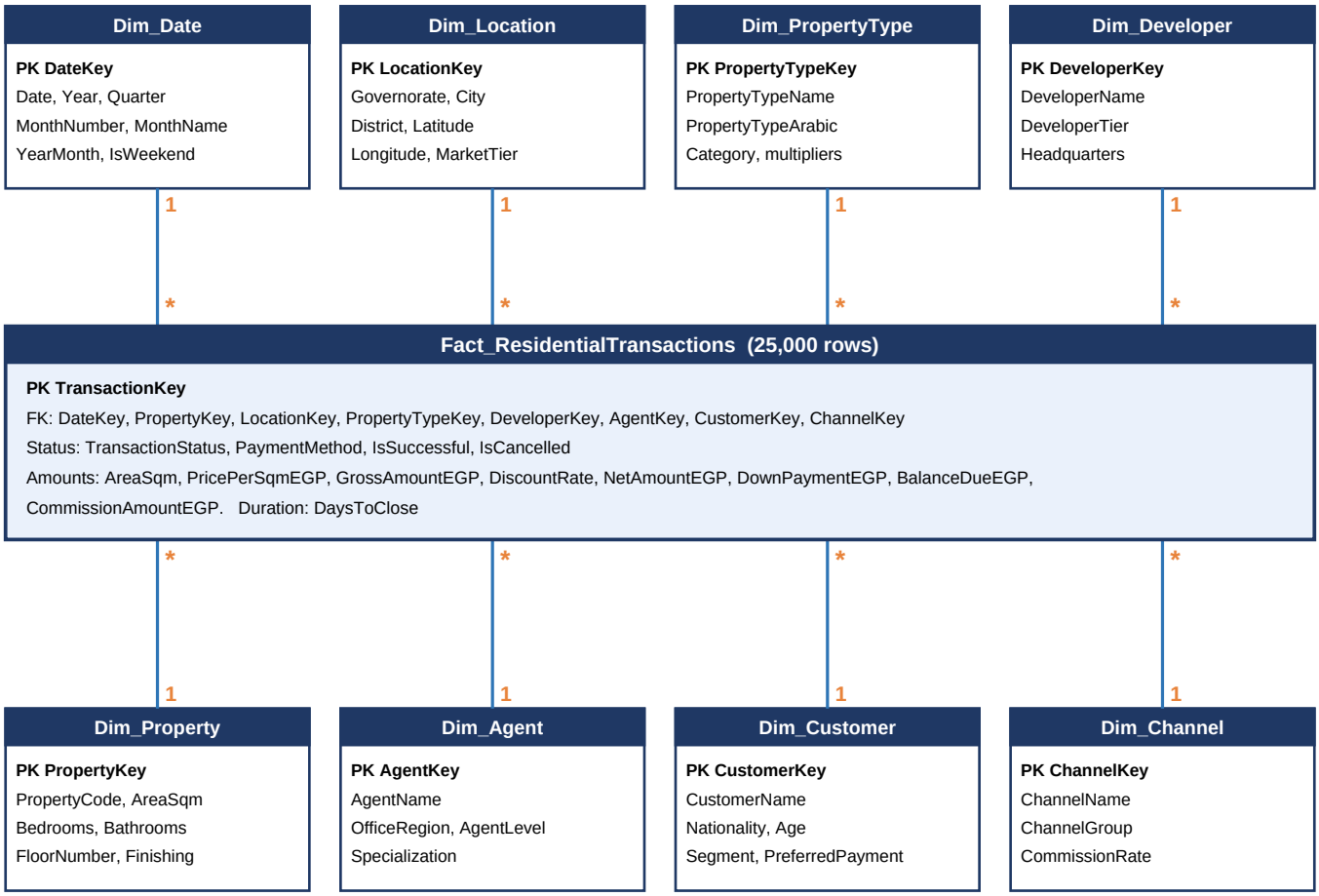
Team: Saleh, Maged, Abd Elkhalek, Sama Raed, Sama Abdelmeged, Mostafa. These two parts correspond to items 2 and 4 of the official "System Analysis & Design" section (DEPI deadline 6 Nov 2026). Status: **proposed, for team review**. Prepared by Saleh from the actual source files; data model decisions to be confirmed by Sama Raed, report design by Abd Elkhalek and Mostafa.

4.2 Database Design & Data Modeling

Overview. The model is a star schema with one fact table and eight dimension tables, loaded from nine CSV files (the README describes the data as synthetic but internally consistent, not an official market extract). The fact grain is one residential transaction or booking for one property on one date. All amounts are in Egyptian pounds (EGP). Row counts below were counted directly from the files.

ER Diagram (star schema)

*1 = one side, * = many side. Single-direction filter from each dimension to the fact.*



Logical schema

Table	Role	Primary key	Foreign keys	Rows	Columns
Fact_ResidentialTransactions	Fact	TransactionKey	DateKey, PropertyKey, LocationKey, PropertyTypeKey, DeveloperKey, AgentKey, CustomerKey, ChannelKey	25,000	22
Dim_Agent	Dimension	AgentKey	-	120	5
Dim_Channel	Dimension	ChannelKey	-	5	4
Dim_Customer	Dimension	CustomerKey	-	2,500	6
Dim_Date	Dimension	DateKey	-	1,461	8
Dim_Developer	Dimension	DeveloperKey	-	20	4
Dim_Location	Dimension	LocationKey	-	30	9
Dim_Property	Dimension	PropertyKey	LocationKey, PropertyTypeKey, DeveloperKey (attributes)	3,000	11
Dim_PropertyType	Dimension	PropertyTypeKey	-	7	7

Relationships

#	From (one side)	To (many side)	Cardinality	Filter direction	Orphans checked
1	Dim_Date[DateKey]	Fact_ResidentialTransactions[DateKey]	One to many	Single (dimension to fact)	0 orphan keys
2	Dim_Property[PropertyKey]	Fact_ResidentialTransactions[PropertyKey]	One to many	Single (dimension to fact)	0 orphan keys
3	Dim_Location[LocationKey]	Fact_ResidentialTransactions[LocationKey]	One to many	Single (dimension to fact)	0 orphan keys
4	Dim_PropertyType[PropertyTypeKey]	Fact_ResidentialTransactions[PropertyTypeKey]	One to many	Single (dimension to fact)	0 orphan keys
5	Dim_Developer[DeveloperKey]	Fact_ResidentialTransactions[DeveloperKey]	One to many	Single (dimension to fact)	0 orphan keys
6	Dim_Agent[AgentKey]	Fact_ResidentialTransactions[AgentKey]	One to many	Single (dimension to fact)	0 orphan keys
7	Dim_Customer[CustomerKey]	Fact_ResidentialTransactions[CustomerKey]	One to many	Single (dimension to fact)	0 orphan keys
8	Dim_Channel[ChannelKey]	Fact_ResidentialTransactions[ChannelKey]	One to many	Single (dimension to fact)	0 orphan keys

The "0 orphan keys" result was recalculated from the CSV files: every fact foreign key exists in its dimension, and every dimension key is unique. This was checked in the data files, not yet in the Power BI model view.

Physical schema (tables, columns, Power BI data types)

Types are the Power BI types applied in Power Query (see the Power Query M code). Column names and order are identical to the source files; Power Query adds, removes and renames no columns.

Fact_ResidentialTransactions (25,000 rows, 22 columns)

Column	Power BI type	Key / notes
TransactionKey	Whole number	PK. Grain: one transaction/booking for one property on one date
DateKey	Whole number	FK to Dim_Date[DateKey]
PropertyKey	Whole number	FK to Dim_Property[PropertyKey]
LocationKey	Whole number	FK to Dim_Location[LocationKey]
PropertyTypeKey	Whole number	FK to Dim_PropertyType[PropertyTypeKey]
DeveloperKey	Whole number	FK to Dim_Developer[DeveloperKey]
AgentKey	Whole number	FK to Dim_Agent[AgentKey]
CustomerKey	Whole number	FK to Dim_Customer[CustomerKey]
ChannelKey	Whole number	FK to Dim_Channel[ChannelKey]
TransactionStatus	Text	Completed / Reserved / Pending / Cancelled
PaymentMethod	Text	Cash / Installment Plan / Mortgage
AreaSqm	Decimal number	
PricePerSqmEGP	Decimal number	
GrossAmountEGP	Decimal number	= AreaSqm x PricePerSqmEGP; kept on cancelled rows
DiscountRate	Decimal number	
NetAmountEGP	Decimal number	= Gross x (1 - DiscountRate); 0 if cancelled
DownPaymentEGP	Decimal number	
BalanceDueEGP	Decimal number	= Net - DownPaymentEGP
DaysToClose	Whole number	0 for Pending and Cancelled (structural)
IsSuccessful	True/False	True for Completed rows
IsCancelled	True/False	True for Cancelled rows
CommissionAmountEGP	Decimal number	Completed rows only: Net x channel rate

Dim_Agent (120 rows, 5 columns)

Column	Power BI type	Key / notes
AgentKey	Whole number	PK, unique
AgentName	Text	
OfficeRegion	Text	
AgentLevel	Text	
Specialization	Text	

Dim_Channel (5 rows, 4 columns)

Column	Power BI type	Key / notes
ChannelKey	Whole number	PK, unique
ChannelName	Text	
ChannelGroup	Text	
CommissionRate	Decimal number	Fraction applied to completed net amount

Dim_Customer (2,500 rows, 6 columns)

Column	Power BI type	Key / notes
CustomerKey	Whole number	PK, unique
CustomerName	Text	
Nationality	Text	
Age	Whole number	
CustomerSegment	Text	
PreferredPaymentMethod	Text	

Dim_Date (1,461 rows, 8 columns)

Column	Power BI type	Key / notes
DateKey	Whole number	PK. Integer yyyymmdd
Date	Date	Mark as the date table; 1,461 continuous days 2022-01-01 to 2025-12-31
Year	Whole number	
Quarter	Text	
MonthNumber	Whole number	
MonthName	Text	
YearMonth	Text	
IsWeekend	True/False	True for Sat/Sun (open decision DQ-04)

Dim_Developer (20 rows, 4 columns)

Column	Power BI type	Key / notes
DeveloperKey	Whole number	PK, unique
DeveloperName	Text	
DeveloperTier	Text	
Headquarters	Text	

Dim_Location (30 rows, 9 columns)

Column	Power BI type	Key / notes
LocationKey	Whole number	PK, unique
Governorate	Text	"6th of October" appears under Cairo and Giza (open decision DQ-02)
City	Text	1 corrupted value (LocationKey 10) repaired in Power Query (DQ-01)
District	Text	
Latitude	Decimal number	
Longitude	Decimal number	
BasePricePerSqm	Whole number	Price assumption (README)
DemandIndex	Decimal number	Demand assumption (README)
MarketTier	Text	

Dim_Property (3,000 rows, 11 columns)

Column	Power BI type	Key / notes
PropertyKey	Whole number	PK, unique
LocationKey	Whole number	Attribute copy; verified equal to the fact for all 25,000 rows. Do not relate to Dim_Location
PropertyTypeKey	Whole number	Attribute copy; same rule. Do not relate to Dim_PropertyType
DeveloperKey	Whole number	Attribute copy; same rule. Do not relate to Dim_Developer
PropertyCode	Text	Pattern EG--; unique
AreaSqm	Decimal number	
Bedrooms	Whole number	
Bathrooms	Whole number	
FloorNumber	Whole number	
DeliveryStatus	Text	
FinishingLevel	Text	

Dim_PropertyType (7 rows, 7 columns)

Column	Power BI type	Key / notes
PropertyTypeKey	Whole number	PK, unique
PropertyTypeName	Text	
PropertyTypeArabic	Text	Arabic text; load as UTF-8
Category	Text	
PriceMultiplier	Decimal number	
SizeMultiplier	Decimal number	
FamilySuitable	True/False	

Normalization and modeling considerations

- **Star over snowflake.** Dimensions are kept flat (for example Dim_Location holds Governorate, City and District in one 30-row table) so every dimension joins directly to the fact. This keeps relationships simple and filters unambiguous.
- **Intentional redundancy in Dim_Property.** It carries LocationKey, PropertyTypeKey and DeveloperKey, and the fact table carries the same three keys. All 25,000 fact rows agree with Dim_Property on these keys and on AreaSqm. Relate only the fact to Location, PropertyType and Developer. Do not also relate Dim_Property to those dimensions, which would create ambiguous filter paths.
- **Stored derived columns.** GrossAmountEGP, NetAmountEGP, BalanceDueEGP and CommissionAmountEGP are derivable but are stored in the source and match their formulas for all 25,000 rows. They are loaded as they are, and DAX should use them rather than recompute them.
- **Status semantics.** Cancelled rows keep gross value and area but have zero net, down payment, balance and commission. Pending and Reserved rows already carry a positive net amount. Measures must filter on TransactionStatus (for example Completed) to report realized sales.
- **Keys and dates.** Keys are whole numbers. DateKey is an integer in yyyymmdd form. Dim_Date should be marked as the date table (1,461 consecutive days). CustomerName is not an identifier (252 distinct names over 2,500 customers), so always use CustomerKey.
- **Open decisions that touch the model:** city and governorate pairing for "6th of October" (DQ-02), the weekend definition in Dim_Date (DQ-04). Both are listed in the data-quality log for team approval.

4.4 UI/UX Design & Prototyping

Design goals. The report must let sales managers see overall performance at once (US-01), compare markets, agents, channels and developers (US-02 to US-07), follow finance figures (US-08), filter by year, month, location, property type and status (US-09), drill from summaries to transactions (US-10) and navigate without training (US-11, NFR-07). Pages must respond in under 5 seconds (NFR-01).

Page map

Page	Purpose	Proposed visuals	Stories / requirements
Overview	Overall performance at a glance	KPI cards (transactions, net sales, average price per sqm, cancellation rate, commission), monthly trend line, top locations bar, status split	US-01, FR-06, FR-07
Locations	Compare markets	Map, net sales by governorate, city/district ranking table, price per sqm by market tier	US-02, US-06
Property types	Price and demand by type	Bar by property type, price per sqm by finishing level, area distribution	US-06
Developers	Partner performance	Sales and cancellation rate by developer and developer tier	US-07
Agents	Agent and office results	Ranked agent table, sales by office region and agent level, commission, days to close	US-03, US-04
Channels	Channel effectiveness	Sales and commission rate by channel, customer segment and nationality mix	US-05
Finance	Revenue and receivables	Gross vs net, discount, down payment, balance due, payment method split	US-08
Details	Transaction list (drill-through)	Table with all key columns, opened from any page, with a back button	US-10, FR-09

The visuals are proposals derived from the stakeholder needs and functional requirements in this document. Abd Elkhalek (design) and Mostafa (interactivity) decide the final layout.

Wireframes (low fidelity)

Figure 1. Overview page: KPI cards on top, trend and rankings below. Left column holds navigation buttons.

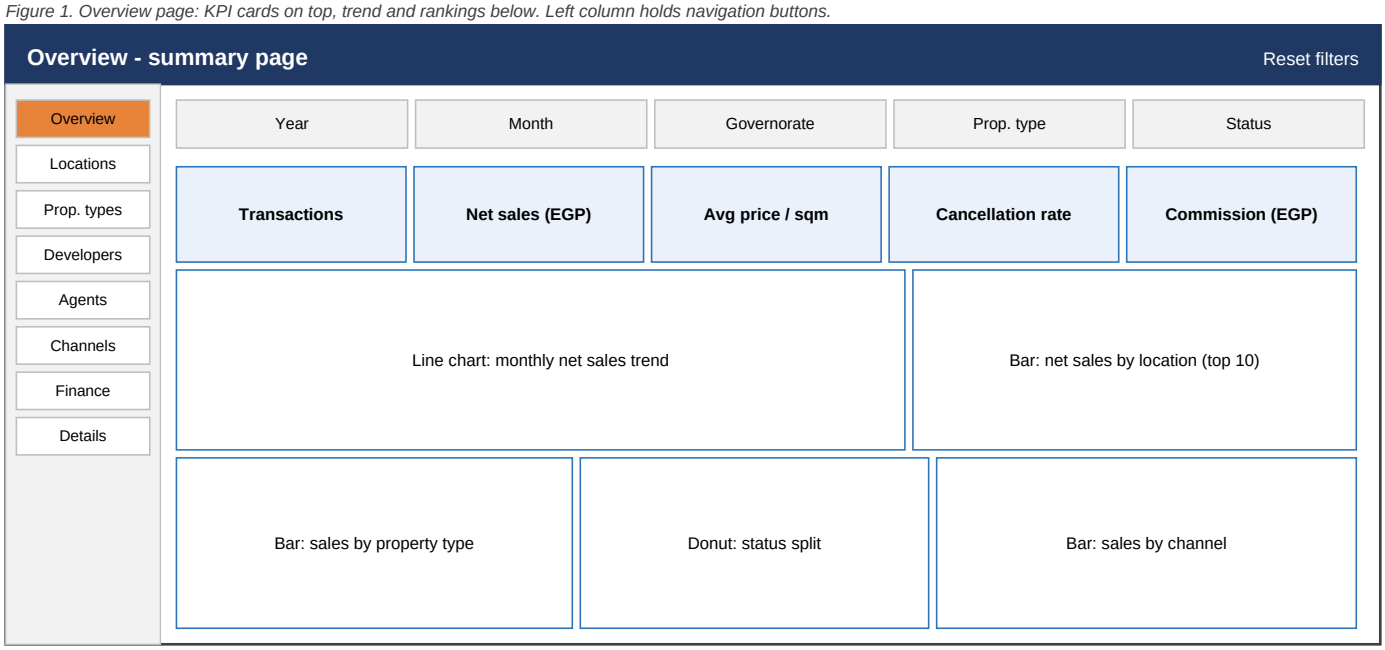


Figure 2. Locations page: map on the left, rankings on the right.

Figure 3. Agents page: ranked table with region and level breakdowns.

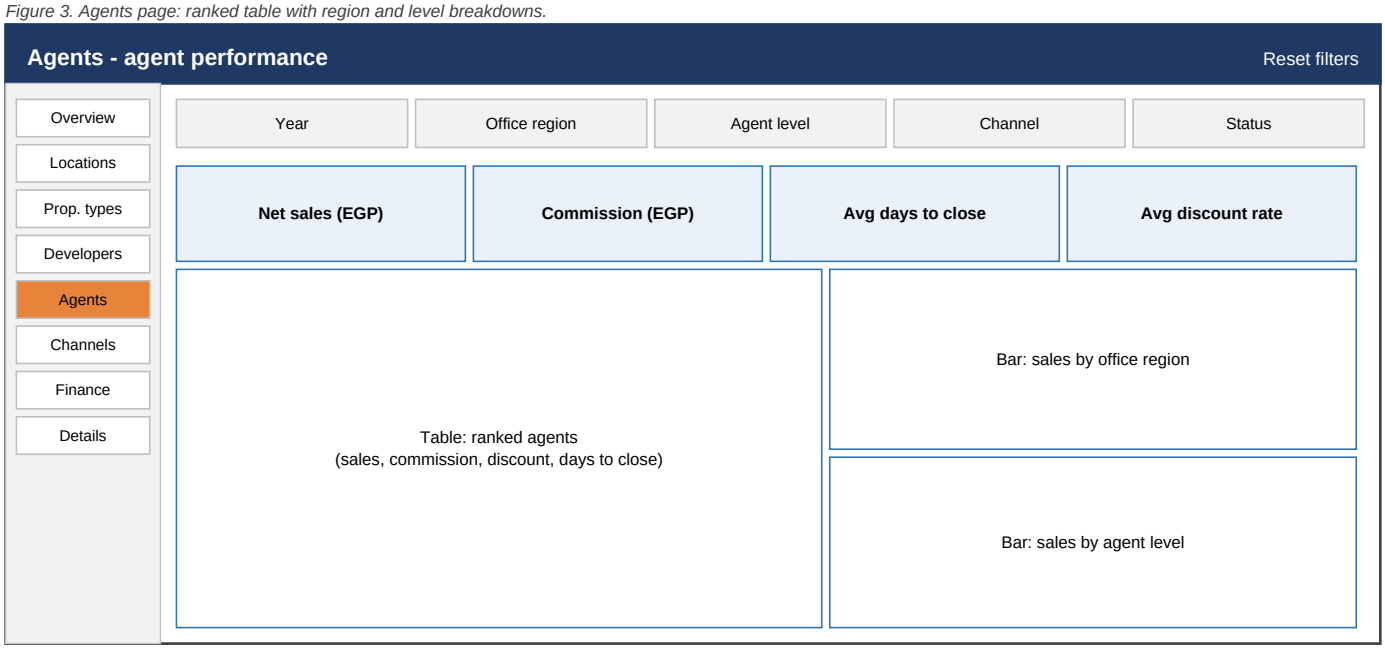
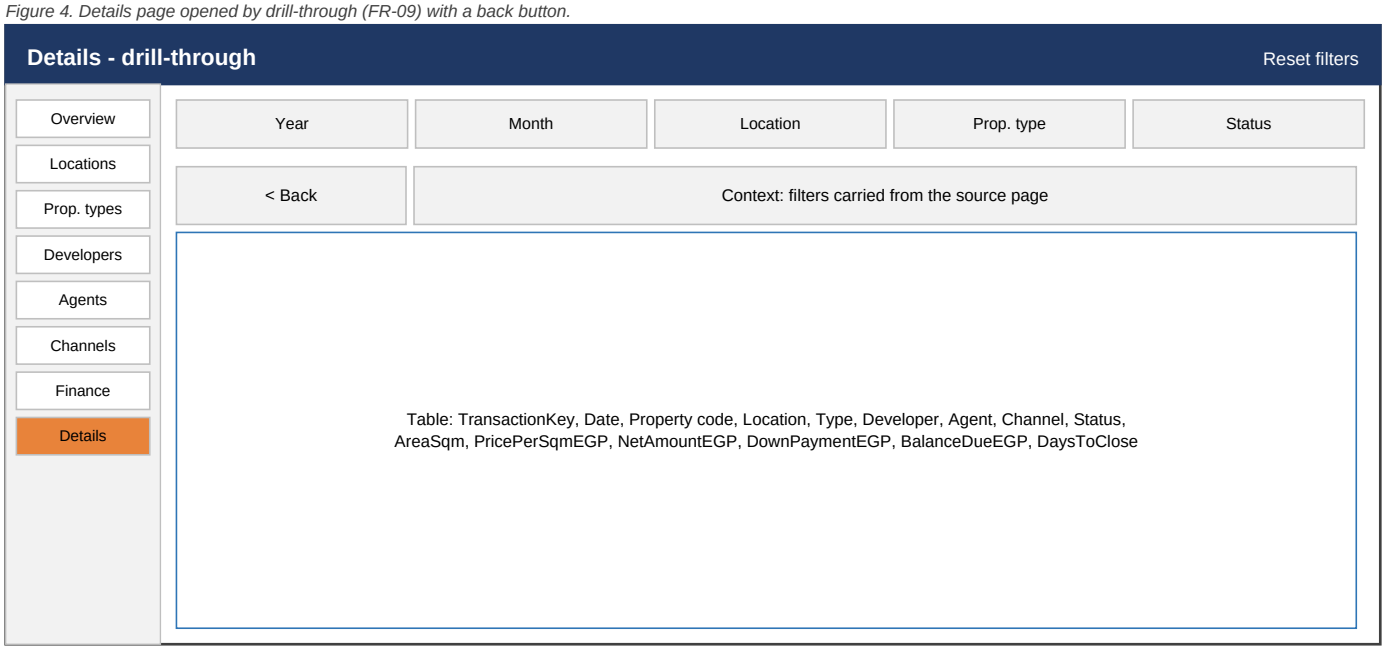


Figure 4. Details page opened by drill-through (FR-09) with a back button.



The remaining pages (Property types, Developers, Channels, Finance) reuse the same grid: slicer row on top, KPI cards, then two to four charts. Navigation buttons stay in the same place on every page.

UI/UX guidelines

Colour	Use	Hex	Contrast with white text
	Primary: titles, header bars, table headers	#1F3864	11.6:1
	Charts: main series, links	#2E75B6	4.8:1
	Charts: second series	#137A6D	5.2:1
	Highlight: selected page button, key callouts (use dark text on it)	#E8833A	2.7:1 (low, so use dark text: 7.7:1)
	Status: Completed	#2E7D4F	5.0:1
	Status: Pending	#8F5F0A	5.5:1
	Status: Cancelled	#C0392B	5.4:1
	Page and card background	#F2F2F2	Use dark grey text #404040 (9.3:1)

Contrast ratios were calculated with the WCAG formula for the exact colours above; 4.5:1 is the usual minimum for normal text, so all colours except orange can carry white text. Reserved status can use the blue #2E75B6.

Area	Guideline
Layout	One grid for all pages: header bar, slicer row, KPI cards on top, charts below, navigation at the left. Keep the same position for the reset-filters and back buttons on every page.
Typography	Segoe UI (Power BI default). Page title 16-18 pt bold, card values 24-28 pt, card labels 10-11 pt, axis and legend text at least 10 pt. Proposed sizes, to be tested on the final canvas.
Colour use	Navy for structure, blue and teal for data, orange for highlights only. Use the status colours above consistently on every page. Never rely on colour alone: add data labels or legends.
Density	Proposed limit of about 8 chart visuals per page, excluding KPI cards, slicers and buttons, to protect the 5-second response target (NFR-01).
Interactivity	Synced slicers for Year and Month across pages, drill-through to Details (FR-09), bookmarks and a reset-filters button (FR-10), tooltips on charts (NFR-06).
Accessibility (NFR-08)	Contrast checked as above, alt text on every visual, logical tab order, mobile layout for Overview.
Metric labels	Label cards with their filter: "Completed net sales" instead of "Total net sales", because the README total also includes Pending and Reserved rows (33.4% of net value). Average days to close only for Completed or Reserved rows. See DQ-06, DQ-07 and DQ-11.
Testing	Check every slicer, drill-through and button on each page with the checklist kept by Mostafa; run the accessibility check before the quality review.

Prototype note

Figures 1 to 4 are static wireframes. The working prototype is the Power BI report itself (.pbix), built by Abd Elkhalek and Mostafa from the model above. The other parts of System Analysis & Design (problem statement, use case diagram, software architecture, data flow, sequence, activity, state and class diagrams, technology stack, deployment and component diagrams) are not covered in these pages.